Sahaj Bhadja

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Summary

Product-focused analyst-engineer with 2+ years turning **user behavior data** into **testable hypotheses**, **SQL-driven insights**, and measurable product improvements via dashboards, experimentation, and cross-functional decision support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **analytics dashboards** and data workflows for 500+ users, reducing query latency from **1.2s to 350ms** and improving trust in behavior metrics used for product decisions.
- Implemented **validation frameworks** across 5K+ monthly jobs, increasing data quality and enabling more reliable analysis of user flows, operational bottlenecks, and feature-level outcomes.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Analyzed usage trends from 10K+ daily requests while shipping features, translating product questions into **measurable hypotheses** and iterative improvements to instructor-facing user workflows.
- Improved release workflows via CI/CD and concurrent processing, reducing turnaround time by **60%** and enabling faster experiment cycles for product and engineering stakeholders.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Developed **Python analysis pipelines** over 100K+ trajectory logs, applying descriptive and inferential evaluation methods to quantify system behavior and surface actionable performance insights.
- Produced clear technical summaries and dashboards for cross-functional teams, improving interpretation of experimental results and speeding data-informed model and product iterations.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Built **API and ETL workflows** with OOP, AWS Glue, and S3, improving response time by **30%** and strengthening the reliability of downstream analytical reporting.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito, increasing pipeline efficiency by **68%** and reducing noise in production analytics and incident tracking.

LST Capitals **India**
Web Developer Intern *November 2022 - January 2023*
- Implemented **data capture workflows** across 50K+ records for booking and payments, improving visibility into conversion stages and operational drop-off points.
- Built webhook-based automations and reporting hooks that improved stakeholder visibility, helping teams monitor process health and prioritize product fixes with clearer evidence.

Skills

Analytics: **SQL**, hypothesis framing, A/B-test readiness, KPI definition, descriptive/inferential analysis
Languages: **Python**, JavaScript/TypeScript, Java, SQL, C++
Data & Visualization: dashboard development, workflow instrumentation, metric validation, report automation
Product & Experimentation: user behavior analysis, integrity/safety-adjacent metrics thinking, cross-functional storytelling
Engineering Tooling: Git, CI/CD, unit/integration testing, Flask/FastAPI/Node.js, PostgreSQL/MySQL/MongoDB

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Built an **analytics workspace** across 10+ databases with schema-aware query generation and structured summaries, enabling rapid exploration of user behavior questions and evidence-backed product decisions.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Instrumented full-stack workflows serving 50K+ users and 1M+ transactions, creating visibility into interaction funnels and operational metrics used to prioritize high-impact product improvements.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **metric-ready operational flows** supporting 100+ daily appointments, with monitoring checks that improved retrieval by **2x** and enabled clearer measurement of workflow reliability.